TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression

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Abstract

Objective: Graves disease (GD) causes autoimmune hyperthyroidism; thyroid eye disease (TED) involves orbital inflammation. We compared associations of genetically proxied blood *TSHR* and *IGF1R* expression with disease susceptibility.

Methods: We screened 2,544 druggable genes using Mendelian randomization (MR). Disease datasets comprised Biobank Japan (BBJ) GD, UK Biobank (UKB) hyperthyroidism and three overlapping FinnGen endpoints: GD, Graves ophthalmopathy and strict Graves ophthalmopathy. Secondary colocalization assessed 24 genes. A fixed five-tissue GTEx extension assessed both genes. Orbital RNA sequencing (four TED patients, one control) was descriptive.

Results: Thirteen genes met the BBJ discovery threshold ($P < 0.05/2{,}544$). Higher genetically proxied *TSHR* expression was associated with lower GD odds in BBJ (odds ratio [OR] per approximately one standard deviation 0.12, 95% confidence interval [CI] 0.07–0.21; $P = 1.09\times10^{-14}$) and FinnGen (OR 0.08, 95% CI 0.05–0.12; $P = 3.19\times10^{-32}$), with concordant directions across all five datasets. *CTLA4*, *HLA-DQA2* and *TUBB* also passed the full-screen threshold in both GD datasets. Blood *IGF1R* estimates consistently indicated higher odds, but only BBJ (OR 1.56, 95% CI 1.07–2.28; $P = 0.0212$) and UKB ($P = 0.0117$) reached uncorrected $P < 0.05$. Secondary colocalization support varied across datasets. Adipose *TSHR*–GD associations passed correction; thyroid and fibroblast *IGF1R* estimates did not reach $P < 0.05$. Mean orbital *TSHR* and *IGF1R* transcript levels exceeded the single control.

Conclusions: Associations were more consistently supported for blood *TSHR* than *IGF1R* expression. Secondary regional analyses limit expression-mediated causal interpretation. Population-control comparisons cannot isolate TED susceptibility within GD or predict treatment response.

Keywords: Graves disease; thyroid eye disease; TSHR; IGF1R; Mendelian randomization; colocalization

Introduction

Thyroid eye disease (TED) can cause proptosis, diplopia and visual impairment in patients with Graves disease (GD) [1, 2]. Teprotumumab, which blocks the insulin-like growth factor 1 receptor (IGF-1R), improves clinical outcomes in active TED [3]. Its efficacy raises a related but distinct question: does inherited variation influencing *IGF1R* expression contribute to disease susceptibility in the same way as variation at *TSHR*, the established GD autoantigen and susceptibility locus?

TSHR and IGF-1R interact in experimental models of orbital disease [4, 5, 6]. However, treatment response, inherited gene regulation and gene expression in inflamed tissue address different biological questions. A receptor may be a useful treatment target without equally strong genetic evidence linking its expression to disease onset. Distinguishing these forms of evidence helps clinicians interpret genetic studies of therapeutic targets.

Mendelian randomization (MR) uses variants associated with an exposure to investigate its relationship with the disease phenotype being studied, termed the outcome [7]. Here, the exposure is blood gene expression and the outcomes are disease diagnoses, not responses to treatment. Inherited variants precede disease onset, but MR supports causal interpretation only when its instrumental-variable assumptions hold. Colocalization asks whether expression and disease associations in a genomic region are consistent with the same causal variant [8]. Distinct nearby variants can appear connected because they are inherited together. Thus, an MR association may be highly significant even when the regional signals do not support a shared variant. MR reports odds ratios (ORs), confidence intervals (CIs) and *P* values; colocalization reports model-based posterior probabilities, which are not *P* values.

Previous genetic studies have already implicated *TSHR* in GD and Graves ophthalmopathy. Ji and colleagues integrated blood methylation, expression and protein associations with GD, reporting *TSHR* among the expression-associated genes also supported in a FinnGen ophthalmopathy dataset [9]. Other MR work has examined inflammatory proteins in ophthalmopathy [10], and recent integrative research has investigated *TSHR* across genetic and transcriptomic data in GD and ophthalmopathy [11]. These findings establish a basis for comparing clinically relevant receptors, but do not determine whether evidence for their blood expression is consistently supported across disease definitions under the same analysis framework. In particular, limited evidence for *IGF1R* could reflect imprecision or dependence on particular instruments, while a strong MR association for *TSHR* does not by itself establish that expression and disease share a causal variant. The remaining question is how robustly these two expression traits are associated with GD and related phenotypes, and how far those associations support biological interpretation.

To address this question, we compared genetically proxied blood expression of *TSHR* and *IGF1R* within a screen of genes encoding potentially druggable proteins, using *CTLA4* and other associated genes as context. We used Biobank Japan (BBJ) GD for discovery, compared the original findings with UK Biobank (UKB) hyperthyroidism and FinnGen ophthalmopathy, and subsequently added FinnGen GD and a stricter ophthalmopathy definition. We evaluated effect direction, precision and statistical support across these datasets, then examined dependence on individual instruments, allele frequencies and colocalization assumptions. A later fixed-panel tissue extension evaluated whether these patterns extended beyond the original blood instruments. Our aim was to characterize the consistency and limits of expression-related genetic evidence for the two receptors. The analyses distinguish GD from broader hyperthyroidism and population-controlled ophthalmopathy; they do not directly estimate TED susceptibility among patients who already have GD or predict treatment response.

Methods

Study design and disease datasets

This two-sample MR study used public association summary statistics and followed STROBE-MR reporting guidance [12]. The exposure used whole-blood expression quantitative trait locus (eQTL) statistics from eQTLGen, comprising up to 31,684 participants, predominantly of European ancestry [13]. The original outcome hierarchy comprised BBJ GD, UKB self-reported hyperthyroidism or thyrotoxicosis, and FinnGen R12 Graves ophthalmopathy [14, 15, 16]. We subsequently added FinnGen R12 E4_GRAVES_STRICT and E4_GRAVES_OPHT_STRICT. The five datasets and their case and control counts are shown in Table 1. The UKB source did not establish an exclusively European analysis sample, so ancestry was not inferred from the reference panel used in its source study.

BBJ remained the discovery dataset. FinnGen GD provided a same-disease comparison, UKB a broader-phenotype comparison, and the two ophthalmopathy endpoints sensitivity analyses of a TED-enriched phenotype. The three FinnGen endpoints contain overlapping participants and were not treated as independent replications or combined by meta-analysis. Their controls were population non-cases for the respective endpoint, not GD patients without TED. The term strict follows the FinnGen endpoint name and does not establish perfect diagnostic specificity. The FinnGen extensions, expanded candidate assessment and instrument diagnostics were post hoc; no prospective registration was available.

Expression instruments and MR

From 4,462 potentially druggable genes [17], instruments were selected from significant blood cis-eQTLs ($P < 5\times10^{-8}$) within 1 Mb of the gene. PLINK clumping used a linkage disequilibrium (LD) threshold of $r^2 < 0.001$ within 10 Mb and 503 European participants from 1000 Genomes Phase 3 [18, 19]. This produced 6,135 single-nucleotide polymorphism (SNP)–gene records for 2,544 genes. The minimum F statistic was 29.7; the median variant-specific expression sample size was 28,092, with a range of 232–31,684. Exposure coefficients and standard errors were reconstructed from Z statistics, sample sizes and reference allele frequencies under unit expression variance [20]. Alleles were aligned with TwoSampleMR; ambiguous palindromic or unavailable variants were excluded. UKB linear-model coefficients and standard errors were divided by $\mu(1-\mu)$, where $\mu = 3{,}731/484{,}598$, to approximate a log-odds scale. BBJ and FinnGen coefficients already used log odds.

We used Wald ratios for single-instrument genes and multiplicative random-effects inverse-variance weighting (IVW), with correction for under-dispersion, for multiple instruments [21]. An instrumental variable (IV) is a genetic variant used to represent variation in expression. Results are reported as ORs per approximately one standard deviation higher blood expression, with 95% CIs and two-sided *P* values. This scale describes reconstructed expression effects, not a drug dose or an intervention. Primary discovery required $P < 0.05/2{,}544$ (1.965×10^{-5}), retaining all eligible genes in the denominator even when an outcome estimate was unavailable. The 13 BBJ discoveries were assessed in FinnGen GD using a focused threshold of 0.05/13 and concordant effect direction. A separate full FinnGen screen used 0.05/2,544 and was exploratory. Nominal $P < 0.05$ was reported descriptively and was not equated with discovery significance.

MR interpretation assumes that instruments predict expression, are not associated with exposure–disease confounders, and influence disease only through the modeled exposure. A small *P* value or high PP.H4 cannot by itself establish these assumptions.

Where applicable, sensitivity assessment included Cochran's Q, alternative MR estimators and omission of individual instruments [22, 23]. Previously completed alternative-estimator results for the original three outcomes were retained; they were not represented as new analyses of every endpoint. Post hoc leave-one-out results assessed dependence on individual variants, not independent replication. Single-instrument *TSHR* results could not undergo these omission or multi-instrument pleiotropy tests.

In a post hoc allele-direction audit, we re-extracted rs179252 and rs179247 from the original expression and disease files. Disease effects were aligned to the blood expression-increasing allele, and G-allele correlations were calculated in European and East Asian reference genotypes and independently checked with PLINK. This audit assessed allele alignment and consistency with published thymus findings, rather than thymic mediation (Supplementary Dataset, TSHR_allele_audit).

Secondary regional analysis and robustness assessment

We report MR associations as the main findings and use colocalization to assess their interpretation. This reporting emphasis was adopted during post hoc reappraisal and was not a prospectively registered hierarchy. Colocalization used complete available regional cis-eQTL statistics, including non-significant variants, matched to each disease dataset [8]. The expanded set included all 13 BBJ discoveries, *IGF1R*, nine additional FinnGen GD discoveries and *HLA-G*, which was selected because its MR significance changed after frequency substitution: 24 genes in total. We used coloc.abf with disease as dataset 1 and expression as dataset 2. Posterior probability of hypothesis 4 (PP.H4) measures support for a shared causal variant under the fitted model. PP.H3 instead supports distinct variants. We used PP.H4 ≥ 0.80 as a descriptive strong-support threshold, with priors p1 = p2 = 10^{-4} and p12 = 10^{-5}, and repeated analyses at p12 = 5×10^{-6} and 10^{-6} [24]. All five hypotheses are defined in Supplementary Methods. Low PP.H4 alone does not establish no disease association, and H3 does not prove multiple independent signals.

We evaluated reference and eQTLGen cohort frequencies, distinguishing changes in effect scaling from changes in variant availability and harmonization. The 24-gene colocalization dataset contains 1,440 settings: 24 genes × five outcomes × four frequency/availability scenarios × three priors. For *TSHR*, we additionally examined narrower windows, a common five-outcome variant set, raw regional disease peaks and all available source cis-eQTLs. For newly evaluated candidates, we checked marginal signal coverage, shared instruments and variant aliases. *C4B* and *HLA-G* underwent instrument-level frequency, weighting, heterogeneity and omission checks. Full-precision results are supplied in one Supplementary Dataset workbook. The original BBJ–ophthalmopathy combined filter is retained as a historical descriptive comparison, not promoted to a prospectively specified validation rule or the primary conclusion of this reappraisal.

Tissue-specific extension

We subsequently assessed both genes in a fixed panel of five GTEx v8 tissues: thyroid, cultured fibroblasts, subcutaneous adipose, visceral adipose and whole blood [25]. The extension plan was fixed before retrieving these tissue results, after the blood analysis was known. GTEx participants were predominantly of European ancestry, not exclusively European. Instruments required cis-eQTL $P < 5\times10^{-8}$, F > 10 and European-reference LD $r^2 < 0.001$. Ten planned gene–tissue comparisons against FinnGen GD formed the primary family ($P < 0.005$); 40 comparisons against the other four datasets formed a secondary family ($P < 0.00125$). Unestimable combinations remained in the correction denominators. Effects used the original slopes and standard errors for inverse-normal-transformed expression. Complete regional summaries were not obtained for this extension, so

tissue colocalization was not performed; significant-only pairs were used solely for instrument selection. Details and all planned combinations are in Table S3 and the Supplementary Dataset.

Orbital expression and implementation

An in-house orbital RNA-sequencing series of four TED patients and one control provided descriptive context only. Technical replicates were combined by biological sample, and transcript abundance was displayed without differential-expression significance testing (Figure S1). These observations were not used to select genes or validate the genetic findings. Analyses used R 4.3.3, TwoSampleMR 0.7.4, coloc 5.2.3 and PLINK 1.9, with independent numerical verification in Python. Detailed procedures, three supplementary tables and two supplementary figures accompany the manuscript. No age- or sex-stratified patient analyses, individual-level multivariable regression, Steiger directionality test or multi-signal colocalization was performed.

AI-assisted tools supported code development, numerical checking and drafting; author verification and responsibility are stated in the Declarations.

Results

Discovery across druggable genes

Higher genetically proxied blood *TSHR* expression was consistently associated with lower disease odds, whereas *IGF1R* estimates indicated higher odds with limited statistical support. Of 2,544 instrumented genes, 2,234 were estimable in BBJ, 2,505 in UKB and 2,480 in each FinnGen endpoint. Unavailable estimates were not counted as negative findings. Thirteen genes met the BBJ discovery threshold (Figure 1; Table 3A): *HLA-A*, *HLA-DQA2*, *CTLA4*, *TSHR*, *C4A*, *HSD3B7*, *TUBB*, *VKORC1*, *TNFSF14*, *PRSS36*, *MAPKAPK5*, *PSMB8* and *IFNGR1*.

TSHR and IGF1R across disease datasets

Higher genetically proxied blood *TSHR* expression was associated with lower odds of BBJ GD (OR 0.12, 95% CI 0.07–0.21; $P = 1.09\times10^{-14}$) and FinnGen GD (OR 0.08, 95% CI 0.05–0.12; $P = 3.19\times10^{-32}$). The same direction was observed for UKB hyperthyroidism (OR 0.09, 95% CI 0.06–0.14; $P = 8.77\times10^{-28}$), FinnGen ophthalmopathy (OR 0.10, 95% CI 0.04–0.24; $P = 2.82\times10^{-7}$) and strict ophthalmopathy (OR 0.15, 95% CI 0.06–0.39; $P = 9.02\times10^{-5}$). All five estimates used rs179252 as the single instrument (Table 2; Figure 2). The strict ophthalmopathy association reached $P < 0.05$ but not the full-screen threshold of 1.965×10^{-5}.

The expression-increasing rs179252-G allele was associated with lower odds in all five disease datasets and was positively correlated with rs179247-G in European ($r = 0.978$; $r^2 = 0.957$; $n = 503$) and East Asian ($r = 0.996$; $r^2 = 0.992$; $n = 504$) reference samples. The latter allele also increased blood *TSHR* expression. Allele-aligned recalculation reproduced the original MR estimates (Supplementary Dataset, TSHR_allele_audit).

IGF1R estimates indicated higher odds in all five datasets, but no association met the full-screen threshold. BBJ GD (OR 1.56, 95% CI 1.07–2.28; $P = 0.0212$) and UKB hyperthyroidism (OR 1.35, 95% CI 1.07–1.70; $P = 0.0117$) reached nominal significance, meaning uncorrected $P < 0.05$. The FinnGen estimates did not: GD OR 1.07 (95% CI 0.86–1.33; $P = 0.539$), ophthalmopathy OR 1.41 (95% CI 0.85–2.33; $P = 0.182$) and strict ophthalmopathy OR 1.52 (95% CI 0.92–2.49; $P = 0.0996$). These findings show directional consistency without statistically convincing confirmation in FinnGen. They do not establish absence of an association.

Comparison with other associated genes

Four genes passed the full-screen threshold in both GD datasets: *TSHR*, *CTLA4*, *HLA-DQA2* and *TUBB* (Table 3A). In FinnGen GD, *CTLA4* was associated with lower odds (OR 0.17, 95% CI 0.08–0.36; $P = 2.15\times10^{-6}$), whereas *HLA-DQA2* (OR 1.90, 95% CI 1.64–2.20; $P = 1.41\times10^{-17}$) and *TUBB* (OR 3.28, 95% CI 2.09–5.17; $P = 2.68\times10^{-7}$) were associated with higher odds. *CTLA4*, an established autoimmune locus [26], and *TUBB* had direction-consistent nominal associations across all five datasets. *HLA-DQA2* also showed higher-odds associations in both ophthalmopathy endpoints, but the UKB estimate was non-significant and in the opposite direction.

Using the focused FinnGen GD threshold of $P < 0.05/13$, seven of the 13 BBJ discoveries had direction-consistent associations. The three additional genes were *C4A* ($P = 4.97\times10^{-5}$), *TNFSF14* ($P = 0.00361$) and *IFNGR1* ($P = 0.00330$). *VKORC1* was not estimable in FinnGen GD; the denominator remained 13. Although *HLA-A* had the smallest BBJ *P* value (2.60×10^{-23}), its FinnGen GD association was non-significant ($P = 0.648$). Thus, discovery significance alone did not determine consistency across datasets.

The full FinnGen GD screen identified 13 significant genes, nine outside the BBJ discovery set: *APOM*, *NEU1*, *IGLV2-18*, *IGLV3-12*, *TNF*, *HIST1H4E*, *LTA*, *LTB* and *C4B* (Table 3B). Several shared the same instrument: *APOM* and *NEU1* used rs3101018, and *IGLV2-18* and *IGLV3-12* used rs5759410. These pairs therefore do not represent independent genetic findings. Five major histocompatibility complex (MHC) region genes among the BBJ discoveries were retained in follow-up; membership of this region was not an exclusion criterion.

Instrument and frequency sensitivity

The two-instrument *CTLA4* estimates were largely supported by rs13030124. Omitting it from FinnGen GD left an imprecise estimate ($P = 0.323$), consistent with the dependence previously observed in UKB and original FinnGen ophthalmopathy. Heterogeneity was present in FinnGen GD for *CTLA4* (Q-test $P = 0.0275$), *TUBB* ($P = 3.67\times10^{-5}$) and *C4A* ($P = 1.27\times10^{-12}$). Their cross-dataset associations should therefore be distinguished from agreement among instruments. The original *IGF1R* BBJ and UKB estimates no longer reached nominal significance after excluding rs2654980 ($P = 0.0751$ and 0.468). This indicates sensitivity to instrument removal, which can reflect changes in precision as well as the contribution of the omitted variant; it does not by itself identify an invalid instrument (Figure S2; Supplementary Dataset).

Frequency substitution preserved the original 13 BBJ discoveries and the principal *TSHR* and *IGF1R* interpretation, but not every exploratory FinnGen classification (Table 3C; Table S2). For *C4B* in FinnGen GD, the MR *P* value changed from 1.10×10^{-5} to 1.48×10^{-4}, crossing the full-screen threshold; both versions had Q-test $P < 10^{-6}$. For *HLA-G*, *P* changed from 0.107 to 1.72×10^{-5} as the IVW weight of rs150254595 increased from 21.3% to 58.9%. Omitting this variant left non-significant estimates under either frequency source. Independent calculations reproduced the eight full-set and 72 instrument-omission estimates for these two genes. Precision also varied across the screen: only 14.6% of estimable BBJ genes could detect an OR of 1.5 with 80% power under the earlier normal-approximation benchmark (Supplementary Dataset, Detection_original).

Secondary colocalization findings

Regional analyses provided a further check on expression-mediated interpretation (Figures 2 and 3; Table S1). For *TSHR*, PP.H4 was 0.951 in BBJ, 0.986 in FinnGen ophthalmopathy and 0.891 in strict ophthalmopathy under the primary prior. It was lower in UKB (0.226) and FinnGen GD (0.264), where PP.H3 was 0.774 and 0.736,

respectively. The latter patterns favoured distinct variants under the fitted model despite very small MR *P* values. This discordance persisted in narrower regions and a common five-outcome variant set. The matched analyses retained the regional disease peaks, and all three FinnGen comparisons shared 6,579 variants in the full cis-region. Lowering the shared-variant prior p12 from 10^{-5} to 10^{-6} reduced PP.H4 to 0.661 in BBJ, 0.875 in ophthalmopathy and 0.451 in strict ophthalmopathy.

IGF1R PP.H4 ranged from 0.010 to 0.404. In UKB, expression-only, distinct-variant and shared-variant probabilities were 0.400, 0.196 and 0.404, respectively. *CTLA4* PP.H4 was 0.201 in BBJ and 0.254 in FinnGen GD, compared with 0.953 in UKB and 0.978 and 0.979 in the two ophthalmopathy endpoints. Neither *HLA-DQA2* nor *TUBB* reached PP.H4 of 0.001 in any dataset. No selected gene reached PP.H4 ≥ 0.80 in FinnGen GD; this statement applies to the 24 genes assessed, not all 2,544 screened genes. *TNFSF14* and *IFNGR1* had strong BBJ support (0.994 and 0.989) without consistent support across the other definitions. For the two *IGLV* genes in BBJ, the absent lead expression variant and weak overlapping expression signals limited interpretation of low PP.H4. The nine additional FinnGen discoveries and sensitivity-selected *HLA-G* lacked strong support in every dataset. The chromosome 16p11.2 discoveries remained compatible with a shared regional association rather than three independent expression targets.

Tissue-specific extension

Four of ten gene–tissue combinations provided a qualifying instrument, each retaining one SNP after clumping (F = 32.1–61.1). Higher genetically proxied *TSHR* expression in subcutaneous and visceral adipose was associated with lower FinnGen GD odds (OR 0.36, 95% CI 0.31–0.43; $P = 1.28\times10^{-34}$ and OR 0.50, 95% CI 0.43–0.58; $P = 4.09\times10^{-19}$, respectively), with corrected support in BBJ and UKB. Subcutaneous estimates also passed correction in both ophthalmopathy datasets; visceral estimates did not ($P = 0.0368$ and 0.186). Neither thyroid nor cultured-fibroblast *IGF1R* estimates reached $P < 0.05$ in any dataset. Their FinnGen GD estimates were OR 0.76 (95% CI 0.56–1.02; $P = 0.0692$) and OR 1.11 (95% CI 0.89–1.38; $P = 0.359$). All 20 estimates and 30 unestimable comparisons are reported (Table S3; Supplementary Dataset, Tissue_extension).

Descriptive orbital expression

Mean orbital *TSHR*, *IGF1R* and *CTLA4* transcript levels in four TED samples exceeded those in the single control (Figure S1). With one control and no confirmatory expression testing, these observations do not establish differential expression or explain the genetic associations.

Discussion

This study addresses how consistently blood expression-related genetic evidence supports two clinically relevant receptors across GD and related disease definitions. *TSHR* showed associations with lower disease odds in both GD datasets, with concordant directions in broader hyperthyroidism and ophthalmopathy. *IGF1R* estimates consistently indicated higher odds, but only the BBJ and UKB estimates reached uncorrected $P < 0.05$, and none met the discovery threshold. The contribution is a comparison of the reproducibility and limitations of these associations within the same analytical framework. It does not establish a difference in biological importance between the receptors, and statistical significance in one gene but not another is not a formal test of different effects.

Our findings extend an existing body of genetic evidence. Ji and colleagues had already linked blood *TSHR* expression to GD and reported support in FinnGen ophthalmopathy [9]; recent integrative work has also investigated *TSHR* in GD and ophthalmopathy [11]. We add a focused comparison with *IGF1R*, a GD comparison spanning BBJ and FinnGen, and an assessment of both genes alongside the wider druggable-gene screen. The result is consistent association evidence for *TSHR*, limited evidence for *IGF1R*, and an explicit distinction between strong MR associations and variable regional support for expression-mediated causality. This comparison uses shared public resources, including eQTLGen and FinnGen, so agreement with earlier work is not wholly independent replication. Different MR methods, FinnGen releases and colocalization settings also preclude interpreting differences between reports as direct contradictions. Our additional evidence concerns the consistency and sensitivity of the associations, rather than discovery of a new susceptibility locus.

The other associated genes provide useful context for this central comparison. *CTLA4*, *HLA-DQA2* and *TUBB* accompanied *TSHR* among genes meeting the full-screen threshold in both GD datasets. *CTLA4* also showed direction-consistent associations across all five outcomes, supporting its use as an autoimmune comparator. However, its multi-instrument estimates relied heavily on rs13030124. *HLA-DQA2* was consistent across GD and ophthalmopathy definitions but not UKB hyperthyroidism, while *TUBB* and *C4A* showed substantial instrument heterogeneity. These qualifications matter when interpreting the apparent corroboration. The strongest BBJ association, *HLA-A*, did not persist in FinnGen GD, illustrating why a smaller discovery *P* value does not necessarily identify a more transferable gene-specific association. Shared instruments and complex MHC linkage further prevent counting every associated expression trait as an independent causal target.

The lower-odds direction for *TSHR* need not contradict the pathogenic role of receptor-stimulating antibodies. Blood transcript abundance, tissue-specific expression and receptor activation are different exposures, and genetic regulatory effects can differ across tissues [25]. Our audit linked the blood expression-increasing rs179252-G allele to rs179247-G in both reference populations. The latter is the GD-protective allele associated with greater thymic transcription in published studies [27, 28] and also increased blood expression in eQTLGen. This directional consistency strengthens the plausibility of a central-tolerance explanation, but does not demonstrate that thymic expression mediates the disease association. These linked variants are not independent replications or proven causal regulators. The marked departure of the MR OR from 1 also reflects division by a small reconstructed expression coefficient. It should not be translated into the effect of experimentally increasing *TSHR* expression. The adipose analyses extended the lower-odds direction beyond blood, but they reused disease GWAS data and implicated the same genomic region. They do not establish an adipose-mediated mechanism or provide independent causal replications.

Experimental and clinical studies provide context for relating TSHR-directed immunity to orbital disease. TSHR can undergo cleavage and release its extracellular A-subunit. Expression of this subunit in an adenoviral immunization model induced greater thyroid-stimulating autoimmunity than a minimally cleaving receptor [29]; plasmid immunization subsequently reproduced orbital adipogenesis, fibrosis and muscle pathology in mice [30]. Daser and colleagues further reported enhanced inflammatory and hyaluronan responses to TNFα in orbital fibroblasts from TSHR-immunized mice [31]. Graves ophthalmopathy has also been documented in primarily euthyroid and hypothyroid patients [32], supporting a distinction between thyroid functional status and orbital immune activity. These observations make the ophthalmopathy associations biologically relevant, but they do not show that an A-subunit shed by human orbital fibroblasts, acting alone, causes TED independently of GD. Our instruments do not measure receptor cleavage or antibody activity. The ophthalmopathy comparisons use

population controls and overlapping FinnGen endpoints, so neither their MR estimates nor their higher colocalization probabilities establish a TED-specific effect within GD or a difference between GD and TED.

For *IGF1R*, the defensible conclusion is that blood expression-related genetic evidence was limited in this analysis. Concordant ORs above 1 are worth reporting, but the two nominal associations and the imprecise ophthalmopathy estimates do not establish a confirmed susceptibility effect. The loss of nominal significance after removing rs2654980 further limits confidence in the original estimates, although instrument removal also reduces precision. The fixed-panel extension found no nominal association for thyroid or cultured-fibroblast *IGF1R*, despite qualifying instruments. Limited support was therefore not confined to the blood analyses that could be estimated. However, the confidence intervals, single-instrument design and lack of orbital eQTLs preclude excluding a clinically relevant effect. Cultured fibroblasts are not orbital fibroblasts, and the unestimable tissue combinations are not negative findings. Teprotumumab efficacy [3] and experimental TSHR–IGF-1R crosstalk [5] establish complementary clinical and biological observations; neither can substitute for the missing statistical confirmation of inherited expression-related susceptibility. The results do not demonstrate that *IGF1R* acts only downstream, or that an association would necessarily become detectable using orbital tissue. The small descriptive orbital series cannot resolve either question.

Colocalization serves as a secondary interpretive analysis rather than a substitute for the MR associations. Its findings nevertheless constrain causal wording. Strong *TSHR* associations in UKB and FinnGen GD coexisted with H3-dominant regional patterns, and BBJ and strict ophthalmopathy support weakened under a lower shared-variant prior. Thus, association strength and evidence for a shared causal variant were not interchangeable. Ji and colleagues used a shared-variant prior of 5×10^{-5} and a PP.H4 threshold above 0.50 [9], whereas our primary settings used 10^{-5} and at least 0.80. These differences further discourage comparing binary colocalization labels across studies without considering the model settings. Single-causal-variant assumptions, ancestry differences and signal coverage can affect these posterior probabilities, particularly in complex immune regions. The regional checks excluded a simple missing-peak explanation for *TSHR* but did not establish the correct causal architecture. For *HLA-DQA2* and *TUBB*, strong MR findings without shared-variant support warrant reporting as associated signals rather than validated expression targets. Multi-signal analysis could clarify some discordance [33], but low PP.H4 neither erases a disease association nor by itself proves that the modeled expression has no causal role.

The sensitivity analyses identify which qualifications matter for the central comparison. Frequency substitution preserved the original BBJ discovery set, whereas *C4B* and *HLA-G* changed significance classifications through altered effect scaling and instrument weighting. Instrument omission identified concentrated evidence for *CTLA4* and reduced statistical support for *IGF1R*. These checks reuse the same data and do not supply independent replication. For researchers evaluating TSHR and IGF-1R, the added answer is that blood expression-related associations are more consistently supported for *TSHR*, while its tissue mechanism and the extent of an inherited *IGF1R* contribution remain unresolved. Independent expression instruments and tissue-relevant regulatory data would help address these uncertainties. Comparing GD patients with and without TED, while accounting for follow-up and selection into the GD group, would address orbital susceptibility directly. The present results inform interpretation and follow-up research, but do not guide treatment selection.

Limitations

The principal limitations involve the exposure and phenotype definitions. Predominantly European blood eQTL effects were applied to BBJ, requiring transferability of expression associations across ancestries. UKB self-reported hyperthyroidism is broader than GD, and an exclusively European source sample was not established.

FinnGen's overlapping, population-controlled endpoints do not directly compare GD patients with and without TED; cohort differences and phenotype differences cannot be separated cleanly. Participant overlap with eQTLGen could not be quantified. Cohort-frequency replacement, which uses a release excluding Framingham participants, addresses only part of expression reconstruction. It does not establish correct tissue specificity, remove LD-reference limitations or independently validate rare-variant effects. The orbital series has a single control, no external replication and no inferential role. The GTEx extension had four estimable combinations, each with one instrument, and predominantly European rather than ancestry-homogeneous exposure data. It reused the disease datasets and lacked tissue-specific regional colocalization.

The analytical limitations include post hoc selection, few instruments and a single-causal-variant colocalization model. *TSHR* has one retained European-reference instrument, precluding multi-instrument pleiotropy tests; several other genes have few variants or heterogeneous estimates. MHC analyses are particularly vulnerable to complex LD and correlated expression, and low PP.H4 can reflect model mismatch or limited signal coverage. Multi-signal colocalization, directionality testing and formal between-phenotype effect tests were not performed. Only 24 selected genes underwent colocalization; therefore its negative classifications cannot be generalized to the entire screen. Discovery and focused follow-up thresholds address specified families of tests, while the expanded scans and sensitivity analyses remain exploratory. Neither a non-significant MR result nor a posterior below 0.80 establishes absence of an effect.

Conclusions

Genetically proxied blood *TSHR* expression showed consistent associations with lower odds of GD and related phenotypes, supported by cross-dataset comparisons within a broader screen of druggable genes. The adipose extension supported the same *TSHR* direction. Blood *IGF1R* estimates indicated higher odds but provided limited statistical evidence, and thyroid/fibroblast estimates provided no nominal support. These findings place the strongest emphasis on association; secondary colocalization limits claims that blood expression mediates disease risk. Experimental TSHR autoimmunity provides biological context for orbital involvement, but this study does not establish that pathway or isolate TED susceptibility within GD.

Declarations

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of interest. The authors declare that they have no conflict of interest.

Ethics approval. This study used publicly available, de-identified summary statistics and an institutionally approved orbital tissue dataset. The in-house orbital transcriptomic component was approved by the Institutional Review Board of Pusan National University Hospital (approval number 2104-018-102) and was conducted in accordance with the Declaration of Helsinki.

Informed consent. Written informed consent was obtained from all individual participants included in the in-house orbital tissue study.

Data availability. Instruments, primary MR and colocalization results, and sensitivity results are supplied as Supplementary Dataset. Analysis code is available from the corresponding author on reasonable request. Public summary statistics analyzed in this study are available from their original repositories: blood *cis*-eQTL data from

the eQTLGen Consortium, tissue eQTL data from GTEx v8, the Biobank Japan Graves disease and UK Biobank hyperthyroidism genome-wide association statistics through the GWAS Catalog, and the FinnGen Release 12 GD and ophthalmopathy statistics from FinnGen. The in-house orbital RNA-seq data are available from the corresponding author on reasonable request, subject to institutional and ethical restrictions.

Author contributions. J.P. conceived and designed the study, performed the analyses, and drafted the manuscript. M.-S.K. contributed to data collection and interpretation. K.-H.S. and S.-W.Y. supervised the study and revised the manuscript.

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Table 1. Data sources

Dataset / endpoint	Cases	Controls	Exposure or clinical role
eQTLGen blood expression	—	—	Exposure: up to 31,684; predominantly European
BBJ GD (GCST90018627)	2,809	172,656	Discovery; East Asian
FinnGen GD (E4_GRAVES_STRICT)	3,962	496,386	Post hoc GD comparison; Finnish
UKB hyperthyroidism (GCST90038636)	3,731	480,867	Self-report; ancestry not established as exclusively European
FinnGen GO (GRAVES_OPHT)	858	499,490	Original ophthalmopathy comparison; Finnish
FinnGen GO strict (E4_GRAVES_OPHT_STRICT)	753	499,595	Post hoc definition sensitivity; Finnish

GD, Graves disease; GO, Graves ophthalmopathy; HT, hyperthyroidism/thyrotoxicosis. The three FinnGen R12 endpoints overlap and use population non-cases for their respective definitions. Strict is the source endpoint label, not proof of perfect specificity. The descriptive orbital series (four TED patients, one control) is separate from the genetic datasets.

Table 2. TSHR, IGF1R and CTLA4 across five disease datasets

Gene	Dataset	IVs	OR (95% CI)	P value	PP.H3	PP.H4
TSHR	BBJ GD	1	0.12 (0.07–0.21)	**1.09×10^{-14}**	0.049	**0.951**
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	**3.19×10^{-32}**	0.736	0.264
TSHR	UKB HT	1	0.09 (0.06–0.14)	**8.77×10^{-28}**	0.774	0.226
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	**2.82×10^{-7}**	0.014	**0.986**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	**9.02×10^{-5}**	0.082	**0.891**
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	**0.0212**	0.236	0.073
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.259	0.010
IGF1R	UKB HT	4	1.35 (1.07–1.70)	**0.0117**	0.196	0.404
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.346	0.032
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.391	0.038
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	**5.45×10^{-15}**	0.799	0.201
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	**2.15×10^{-6}**	0.746	0.254
CTLA4	UKB HT	2	0.21 (0.08–0.57)	**0.0022**	0.047	**0.953**
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	**0.0118**	0.022	**0.978**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	**0.00888**	0.021	**0.979**

IVs, instrumental variables (retained genetic variants). Wald ratio for one IV; random-effects IVW for multiple IVs. ORs are per approximately one SD higher reconstructed blood expression. Bold P values denote nominal $P < 0.05$ only. Discovery uses 0.05/2,544. PP.H3 supports distinct variants; PP.H4 supports one shared variant. Bold PP.H4 denotes ≥0.80 under $p12 = 10^{-5}$. These are probabilities, not P values. CTLA4 multi-IV results are concentrated in rs13030124; TSHR is single-IV in all datasets.

Table 3. Screen and supporting analyses — A

BBJ discovery gene	BBJ OR (95% CI)	BBJ P	FinnGen GD OR (95% CI)	FinnGen GD P	Focused support	GD PP.H4
HLA-A	1.98 (1.73–2.27)	**2.60×10^{-23}**	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	2.39 (1.95–2.93)	**8.30×10^{-17}**	1.90 (1.64–2.20)	**1.41×10^{-17}**	Yes	<0.001
CTLA4	0.18 (0.11–0.27)	**5.45×10^{-15}**	0.17 (0.08–0.36)	**2.15×10^{-6}**	Yes	0.254
TSHR	0.12 (0.07–0.21)	**1.09×10^{-14}**	0.08 (0.05–0.12)	**3.19×10^{-32}**	Yes	0.264
C4A	0.45 (0.35–0.57)	**2.23×10^{-10}**	0.41 (0.26–0.63)	**4.97×10^{-5}**	Yes	<0.001
HSD3B7	0.29 (0.18–0.47)	**2.04×10^{-7}**	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	1.49 (1.28–1.74)	**3.14×10^{-7}**	3.28 (2.09–5.17)	**2.68×10^{-7}**	Yes	<0.001
VKORC1	0.13 (0.06–0.29)	**6.35×10^{-7}**	NA	NA	NA	0.011
TNFSF14	0.63 (0.52–0.76)	**1.48×10^{-6}**	0.77 (0.65–0.92)	**0.00361**	Yes	0.303
PRSS36	6.00 (2.87–12.57)	**2.02×10^{-6}**	1.40 (1.00–1.96)	**0.049**	No	0.063
MAPKAPK5	0.977 (0.967–0.987)	**5.28×10^{-6}**	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	1.25 (1.13–1.38)	**6.86×10^{-6}**	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	2.10 (1.51–2.91)	**9.41×10^{-6}**	1.62 (1.18–2.24)	**0.0033**	Yes	0.313

Panel A: focused support requires concordant direction and $P < 0.05/13$; 7 of 13 qualify. NA is unavailable, not a negative result. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.

Table 3. Screen and supporting analyses — B

Additional gene	GD IVs	GD OR (95% CI)	GD P	BBJ PP.H4	GD PP.H4	GO strict PP.H4
APOM	1	0.05 (0.04–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.227
NEU1	1	0.04 (0.03–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.061
IGLV2-18	1	0.56 (0.48–0.66)	**7.52×10^{-12}**	0.029	0.020	0.100
IGLV3-12	1	1.20 (1.14–1.26)	**7.52×10^{-12}**	0.019	0.022	0.101
TNF	1	0.60 (0.51–0.71)	**4.65×10^{-10}**	<0.001	<0.001	<0.001
HIST1H4E	1	2.54 (1.88–3.43)	**1.51×10^{-9}**	<0.001	<0.001	0.007
LTA	1	0.46 (0.35–0.61)	**3.75×10^{-8}**	<0.001	<0.001	<0.001
LTB	1	0.18 (0.10–0.36)	**4.89×10^{-7}**	<0.001	<0.001	<0.001
C4B	2	1.63 (1.31–2.03)	**1.10×10^{-5}**	<0.001	<0.001	0.311

nine additional reference-frequency FinnGen GD discoveries at $P < 0.05/2{,}544$; post hoc and not independent of the full screen. APOM/NEU1 share rs3101018; IGLV2-18/IGLV3-12 share rs5759410. The lead IGLV expression variant is absent in BBJ, weakening interpretation of its low PP.H4.

Table 3. Screen and supporting analyses — C

Gene / GD analysis	Reference OR (95% CI)	Reference P	Cohort-frequency OR (95% CI)	Cohort P	Interpretation
C4B	1.63 (1.31–2.03)	**1.10×10^{-5}**	1.75 (1.31–2.33)	**1.48×10^{-4}**	Heterogeneous; loses screen significance
HLA-G	0.86 (0.72–1.03)	0.107	0.60 (0.47–0.75)	**1.72×10^{-5}**	Weight-sensitive; single-variant dependence

HLA-G was added because of frequency sensitivity, not primary discovery. Neither C4B nor HLA-G gains strong GD colocalization. All 24 genes and five datasets are available in the workbook. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.

Figure 1. Study design and BBJ discovery

A Blood expression instruments
4,462 druggable genes considered
2,544 genes; 6,135 SNP–gene records

→

BBJ GD discovery
2,234 estimable genes; 13 discoveries
Threshold P < 0.05/2,544

→

Disease-definition comparisons
FinnGen GD, UKB hyperthyroidism
Two overlapping GO definitions

Regional follow-up: TSHR and IGF1R at the centre; CTLA4 comparator; 24 selected genes in total.

B BBJ gene-level MR associations

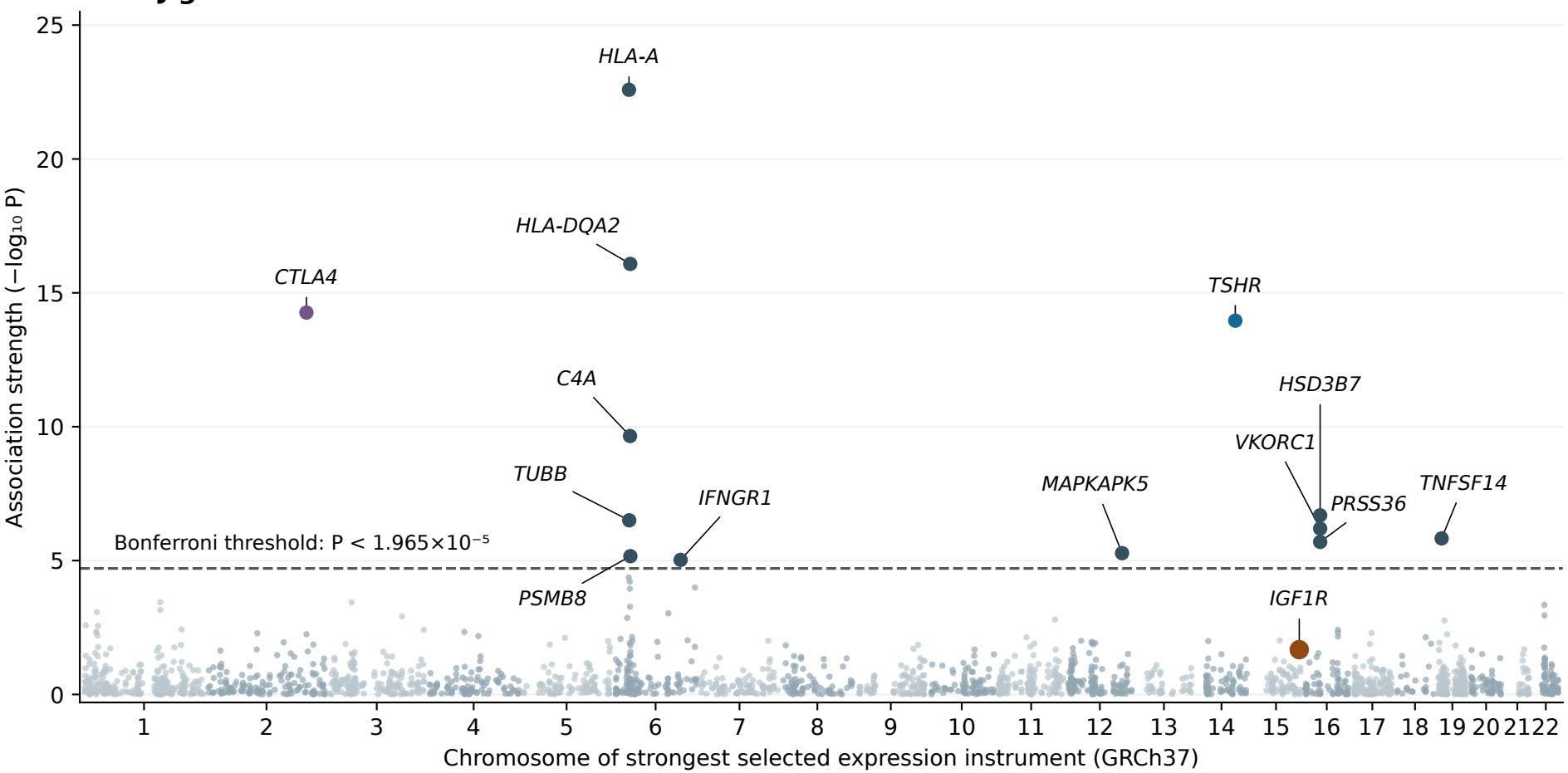


Figure 2. TSHR and IGF1R across disease definitions, with CTLA4 as comparator

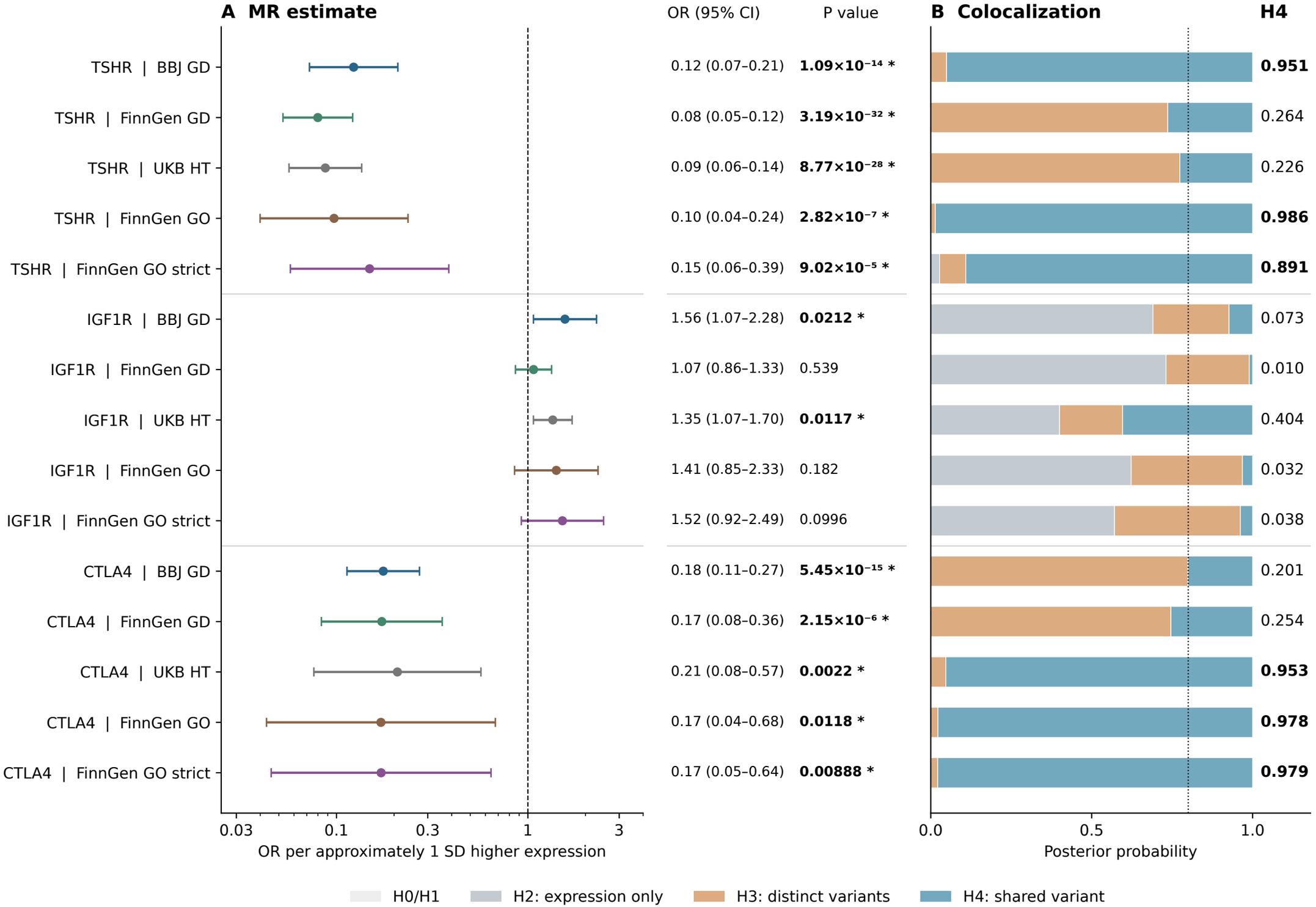
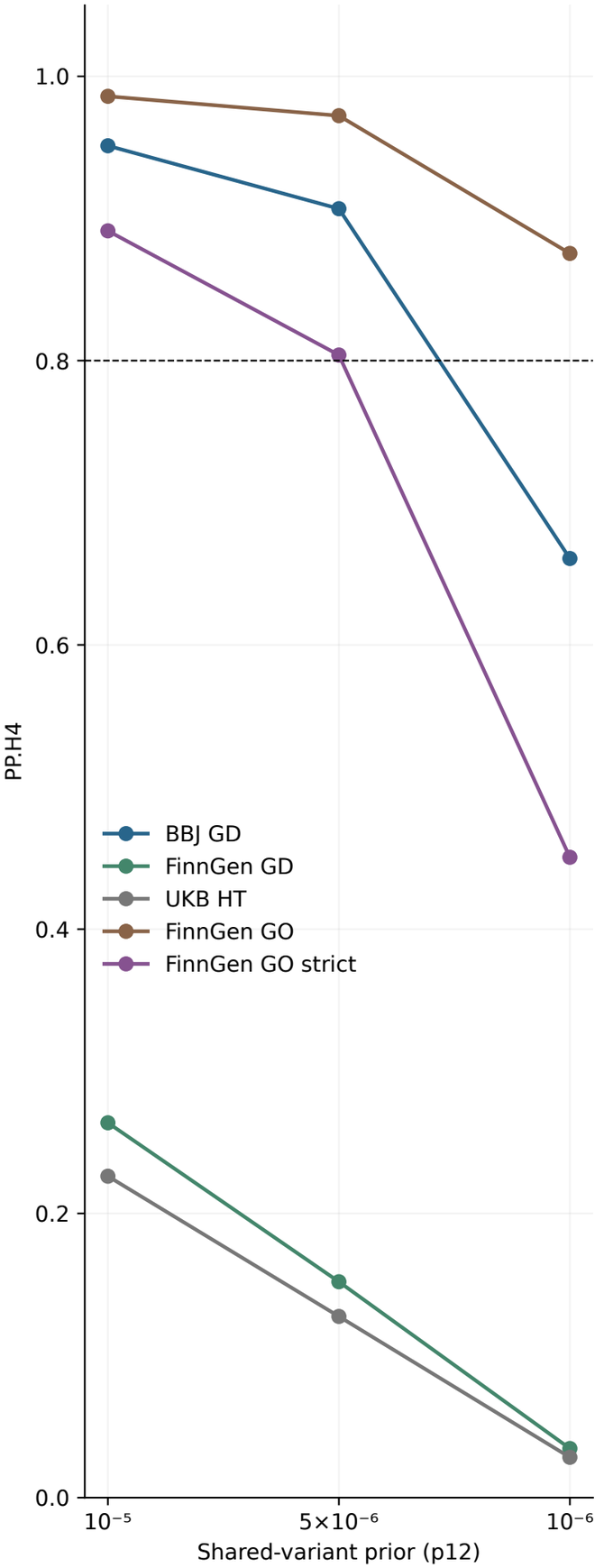


Figure 3. Regional genetic evidence and TSHR prior sensitivity

A Shared-variant probability (PP.H4)

	BBJ GD	FinnGen GD	UKB HT	FinnGen GO	FinnGen GO strict
TSHR	0.951	0.264	0.226	0.986	0.891
IGF1R	0.073	0.010	0.404	0.032	0.038
CTLA4	0.201	0.254	0.953	0.978	0.979
HLA-A	<0.001	<0.001	<0.001	<0.001	<0.001
HLA-DQA2	<0.001	<0.001	<0.001	<0.001	<0.001
C4A	<0.001	<0.001	<0.001	0.030	0.326
HSD3B7	0.636	0.011	0.095	0.026	0.046
TUBB	<0.001	<0.001	<0.001	<0.001	<0.001
VKORC1	0.388	0.011	0.030	0.030	0.048
TNFSF14	0.994	0.303	0.237	0.017	0.018
PRSS36	0.168	0.063	0.115	0.037	0.033
MAPKAPK5	<0.001	0.041	0.005	0.026	0.029
PSMB8	<0.001	<0.001	<0.001	<0.001	<0.001
IFNGR1	0.989	0.313	0.052	0.020	0.021
APOM	<0.001	<0.001	<0.001	0.006	0.227
NEU1	<0.001	<0.001	<0.001	0.003	0.061
IGLV2-18	0.029	0.020	0.024	0.082	0.100
IGLV3-12	0.019	0.022	0.024	0.083	0.101
TNF	<0.001	<0.001	<0.001	<0.001	<0.001
HIST1H4E	<0.001	<0.001	<0.001	0.004	0.007
LTA	<0.001	<0.001	<0.001	<0.001	<0.001
LTB	<0.001	<0.001	<0.001	<0.001	<0.001
C4B	<0.001	<0.001	<0.001	0.030	0.311
HLA-G	<0.001	<0.001	<0.001	<0.001	0.001

B TSHR prior sensitivity



Supplementary material

Detailed methods

Dataset definitions and analysis chronology

BBJ GD (GCST90018627) and UKB self-reported hyperthyroidism/thyrotoxicosis (GCST90038636) retained the source sample sizes and estimates. UKB used BOLT-LMM linear-model statistics; its report included adjustment for sex, age, BMI, assessment centre, ethnicity, genotyping batch and 20 principal components. The available documentation did not establish an exclusively European analysis sample. FinnGen R12 GWAS counts were 858/499,490 for GRAVES_OPHT, 3,962/496,386 for E4_GRAVES_STRICT and 753/499,595 for E4_GRAVES_OPHT_STRICT. Counts refer to the analyzed GWAS files, not later phenotype-browser totals. Strict GD combines specified GD/ophthalmopathy diagnostic criteria with the Graves condition and exclusion of amiodarone exposure; strict ophthalmopathy requires its specified ophthalmopathy and thyrotoxicosis codes. The exact endpoint definitions and source manifests are retained with the analysis verification records. None supplies GD-only controls.

The original three-outcome analysis preceded the two FinnGen extensions. The expanded plan was frozen before those new analyses, but was not a prospective registration of the complete study. The 24 colocalization genes comprise 13 BBJ discoveries, IGF1R, nine additional primary FinnGen GD discoveries, and frequency-sensitive HLA-G. No new discovery threshold was optimized from these results.

Instrument selection and effect reconstruction

Significant cis-eQTL instruments ($P < 5\times10^{-8}$ within ±1 Mb of the gene) were clumped at $r^2 < 0.001$ within 10 Mb using the European 1000 Genomes Phase 3 reference (n = 503). Selected instruments and GRCh37 positions are in the Instruments worksheet. $SE = 1/\sqrt{[2f(1-f)(N + Z^2)]}$ and $\beta = Z \times SE$, where f is the assessed-allele frequency and N is the variant-specific expression sample size. Unit expression variance was assumed. Variant $F = Z^2$. The estimate is approximately per expression SD, not per SD of the genetic predictor.

TwoSampleMR harmonise_data(action = 2) aligned alleles and excluded ambiguous palindromic variants with MAF > 0.42. Missing outcome estimates were not imputed. BBJ and FinnGen retained log-odds scales; UKB β and SE were each divided once by $\mu(1-\mu)$, with $\mu = 3{,}731/484{,}598$. GRCh37 expression and GRCh38 FinnGen coordinates were not directly joined: identifiers and allele pairs were checked, with duplicate and alias audits. Source covariate adjustments were retained; patient-level re-adjustment and subgroup analysis were unavailable.

MR estimators and instrument diagnostics

For a single instrument, the Wald coefficient is $\beta Y/\beta X$ and its first-order SE is $SEY/|\beta X|$. Multiple-instrument MR used inverse-variance weighted regression through the origin. Multiplicative random-effects SEs included under-dispersion correction, so heterogeneity could widen the interval. Normal two-sided tests and 95% CIs were used for the primary estimators. The alternative-estimator worksheet preserves weighted median, weighted mode and MR-Egger results for the original three outcomes; mode and Egger intervals use the corresponding t distributions. These alternative estimators were not rerun for all new endpoints.

Post hoc leave-one-out analyses applied IVW when at least two variants remained and the Wald ratio when one remained. Single-instrument TSHR and BBJ CTLA4 were ineligible. Original three-outcome results, new-endpoint results for the fixed 14-gene set, and 72 C4B/HLA-G omission estimates under two frequency sources are retained in separately named worksheets. They reuse the same association data and are not independent replications. C4B/HLA-G instrument weights use $\beta X^2/SEY^2$, normalized within each full model. A single-SNP P value can be much smaller than the full IVW P value when heterogeneity inflates the full-model SE.

Previous TSHR clumping sensitivity used eQTL thresholds of $5\times10^{-8}/5\times10^{-6}$, r^2 thresholds of 0.001/0.01/0.1, and European (n = 503) or East Asian (n = 504) reference panels. Primary settings selected one European and two East Asian variants, and rs179252 was retained in every setting. Previous chromosome 16p11.2 analyses found high East Asian LD among rs4889606, rs34649473 and rs78924645; conditional analysis supported regional dependence rather than three independent gene effects. These earlier clumping and conditional analyses were preserved; the separate allele-direction audit below newly examined rs179252 and rs179247.

TSHR allele alignment and reference LD

In a post hoc audit, original full eQTLGen and disease GWAS files were re-extracted for rs179252 and rs179247. Their GRCh37 positions were 14:81435985 and 14:81432546, and GRCh38 positions were 14:80969641 and 14:80966202. Both non-palindromic allele pairs matched across sources. UKB effects were reversed to the G allele after the existing scale conversion; BBJ and FinnGen required no reversal. Exposure coefficients and first-order Wald estimates were independently reconstructed. Original GWAS P values and recalculated MR P values were retained separately.

G-allele counts at both variants were positively correlated in European (n = 503; r = 0.978322812; r^2 = 0.957115524) and East Asian (n = 504; r = 0.995803919; r^2 = 0.991625445) references, without missing genotypes. Direct BED decoding agreed with PLINK dosage output. PLINK haplotype estimates supported G–G and T–A coupling; these were inferred from unphased genotypes, not directly observed phased haplotypes. Genotype-based r^2 and haplotype-based estimates were kept distinct. Alignment to the G allele resolved the negative default European correlation obtained when counting T at rs179252. The TSHR_allele_audit worksheet retains both eQTL records, both variants across five disease outcomes, all four inferred haplotypes in each population, and MR reproduction. This audit does not identify a causal variant or establish mediation in thymus or orbit.

Colocalization hypotheses and regional inputs

coloc.abf received beta/varbeta data for disease (dataset 1, case-control type) and expression (dataset 2, quantitative type, sdY = 1). Outcome allele frequency was not used in the Bayes factors. Complete available cis-eQTL statistics, not just significant eQTLs, were matched by identifier and allele compatibility. H0 denotes neither trait associated; H1 disease only; H2 expression only; H3 both traits associated through distinct variants; H4 both traits associated through one shared variant. PP.Hk is the posterior probability of hypothesis k, conditional on the model and priors. Dataset order is essential to interpreting H1 and H2.

Primary priors were $p1 = p2 = 10^{-4}$ and $p12 = 10^{-5}$; sensitivity p12 values were 5×10^{-6} and 10^{-6}. The PP.H4 ≥ 0.80 threshold was a reporting convention, not a significance test. H0–H4 sum to one apart from rounding. Values displayed as <0.001 are nonzero full-precision probabilities in the workbook. The top_snp is conditional on H4, not necessarily the disease lead variant. H3 is a competing distinct-variant model, not a count of causal signals. Multi-signal colocalization was not performed.

Four frequency/availability scenarios were used: original_reference (original available reference frequencies), paired_reference and paired_eqtlgen (the same variants with the respective frequencies), and available_eqtlgen (all matching variants with valid cohort frequencies). There are 1,440 gene–outcome–scenario–prior rows. The additional TSHR analysis comprises four windows × five outcomes × four scenarios × three priors = 240 rows. Windows were the full source cis-region, ±500 kb and ±250 kb around the recorded GRCh37 GenePos anchor, and the five-outcome variant intersection. The full cis-source contained 7,309 eQTL rows; all three FinnGen full-region analyses matched 6,579 variants, and the five-outcome intersection contained 4,030. Raw outcome regional scans checked whether the strongest disease variants survived matching. New candidate coverage checks also examined marginal expression strength; the absent rs5759410 lead in BBJ was checked against variant aliases without finding a replacement.

Frequency substitution and precision benchmarks

The published eQTLGen frequency release covered 26,609 participants, excluding Framingham. Reconstructing βX and SEX with fixed Z and N preserves their ratio up to floating-point precision. For a single-IV Wald estimate, frequency scaling changes the effect unit and SE together and preserves the normal-test P value; for IVW, variant weights can change. The new-endpoint MR dataset separates identical-variant scaling from repeat allele harmonization. C4B and HLA-G were audited with identical retained variants and outcome effects, so their differences arise from scaling and weighting rather than a changed variant set. Large frequency discrepancies are documented, not assumed to be resolved by correct identifier matching.

The earlier detection benchmarks use $(z_{1-\alpha/2} + z_{0.80}) \times SE$ as the approximate 80% minimum detectable absolute log-OR. Alpha was 0.05/2,544 for BBJ and 0.05 for the original UKB/FinnGen comparisons. These are precision summaries using observed SEs, not prospective sample-size calculations, post hoc proof of power, or power of colocalization/the combined filter. The Detection_original worksheet retains this original three-outcome scope; no unperformed new-endpoint power analysis is implied.

Fixed-panel tissue extension

The post hoc tissue extension retained a frozen panel of TSHR and IGF1R in five standard-release GTEx v8 tissues. QTL sample counts were 574 (thyroid), 483 (cultured fibroblasts), 581 (subcutaneous adipose), 469 (visceral adipose) and 670 (whole blood). The original v8 archive supplied significant variant–gene pairs and eGenes records for all tested genes; gene-specific release thresholds exceeded 5×10^{-8} for every eligible combination, permitting complete selection at the fixed threshold. Absence from the tested-gene file was distinguished from a minimum eQTL P value above the threshold. The standard release includes multiple ancestries; the overall donor cohort was 85.3% European American, not a tissue-specific percentage.

Variants were restricted to ±1 Mb of the source GENCODE v26 TSS, $P < 5\times10^{-8}$, $F > 10$ and MAF ≥ 0.01. GTEx variant IDs identify GRCh38 REF and ALT, with slope referring to ALT. Actual effect-allele frequencies were reconstructed from each selected eGene MAF and ref_factor, not by treating MAF as ALT frequency. GTEx mappings linked GRCh38 variants to rsIDs and GRCh37 reference coordinates. PLINK clumping used 503 European reference samples, $r^2 < 0.001$ and a 10-Mb window; an independent genotype-based calculation reproduced selection. One of 20 thyroid IGF1R candidates was absent from the reference, while its strongest candidate was retained. All selected variants were non-palindromic and matched exactly in the five disease datasets; no proxies or relaxed thresholds were used.

Each of the four estimable gene–tissue combinations retained one SNP. Wald estimates used βY/βX and SEY/|βX| with normal two-sided P values and 95% CIs; independent R calculations verified all 20 estimates. The UKB linear-to-log-odds conversion was applied once. ORs represent one unit of inverse-normal-transformed normalized expression; they are not directly comparable to an intervention or to the reconstructed eQTLGen scale. Primary adjusted P = min(1,10P) for FinnGen GD; secondary adjusted P = min(1,40P) for the other datasets. The 30 unestimable cells remained in the planned families. Single-instrument analyses cannot support heterogeneity, leave-one-out or MR-Egger diagnostics. Significant-only eQTL pairs were not used for colocalization; full regional data were not obtained from the requester-pays resource, and no tissue posterior was calculated. Tissue_extension in the single Supplementary Dataset retains all 50 cells, availability, selected and excluded candidates, harmonized alleles and source statistics.

Orbital expression and reproducibility

Four TED patients and one control contributed orbital RNA-sequencing data. Technical replicates were combined by biological sample. Figure S1 retains the previously reviewed sample-level transcript abundance and log2[(mean TED TPM + 0.01)/(control TPM + 0.01)]. One control cannot estimate control-group biological variance reliably, and no differential-expression P value was used. The dataset is descriptive and did not influence candidate selection. Collection and laboratory metadata should be confirmed by the tissue investigators before submission; this reanalysis did not independently reconstruct tissue collection or sequencing procedures.

The workbook preserves exported full-precision analytical outputs, not spreadsheet reimplementations of MR. Readme and Column_dictionary identify each sheet, its scope and its actual columns; no outcome_min_p field is described unless present in the source. Source hashes and numerical checks accompany the review package. Reproduction of a calculation validates implementation, not instrument validity, causal assumptions or clinical generalizability.

Table S1. TSHR regional and prior sensitivity

Dataset	Full cis SNPs	Primary PP.H4	Lower-prior PP.H4	PP.H4 range across windows
BBJ GD	4336	**0.951**	0.661	0.951–0.951
FinnGen GD	6579	0.264	0.035	0.264–0.264
UKB HT	5667	0.226	0.028	0.226–0.226
FinnGen GO	6579	**0.986**	**0.875**	0.986–0.987
FinnGen GO strict	6579	**0.891**	0.451	0.891–0.905

Table S1. TSHR regional and prior sensitivity. Lower prior: p12 = 10^{-6}; p1 = p2 = 10^{-4} throughout. Window range uses the full source cis-region, ±500 kb, ±250 kb and five-outcome intersection at the primary prior. It is not a confidence interval. Full regional disease lead variants were retained. All 240 settings are in TSHR_windows.

Table S2. Frequency-sensitive exploratory candidates

Gene / dataset	Frequency	IVs	OR (95% CI)	P value	Q-test P
C4B / FinnGen GD	Reference	2	1.63 (1.31–2.03)	**1.10×10^{-5}**	7.79×10^{-7}
C4B / FinnGen GD	eQTLGen	2	1.75 (1.31–2.33)	**1.48×10^{-4}**	1.36×10^{-8}
C4B / FinnGen GO strict	Reference	2	1.68 (1.44–1.95)	**2.74×10^{-11}**	0.122
C4B / FinnGen GO strict	eQTLGen	2	1.80 (1.46–2.23)	**6.17×10^{-8}**	0.0587
HLA-G / FinnGen GD	Reference	16	0.86 (0.72–1.03)	0.107	9.98×10^{-24}
HLA-G / FinnGen GD	eQTLGen	16	0.60 (0.47–0.75)	**1.72×10^{-5}**	2.26×10^{-10}
HLA-G / FinnGen GO strict	Reference	16	0.91 (0.76–1.08)	0.267	0.0231
HLA-G / FinnGen GO strict	eQTLGen	16	0.70 (0.53–0.92)	**0.0122**	0.131

Table S2. C4B and HLA-G frequency sensitivity. Variant sets and outcome effects were held fixed. Frequency changes rescaled expression estimates and IVW weights. Q tests heterogeneity, not directional pleiotropy; its P value is not a gene-association P value. All 72 omission estimates and instrument weights are provided in the workbook.

Table S3. Tissue-specific expression and FinnGen GD

Gene / tissue	N	IVs	OR (95% CI)	P value	Adjusted P
TSHR / Thyroid	574	0	No qualifying IV	—	—
IGF1R / Thyroid	574	1	0.76 (0.56–1.02)	0.0692	0.692
TSHR / Cultured fibroblasts	483	0	Not tested	—	—
IGF1R / Cultured fibroblasts	483	1	1.11 (0.89–1.38)	0.3591	1.000
TSHR / Subcutaneous adipose	581	1	0.36 (0.31–0.43)	1.28e-34	1.28e-33
IGF1R / Subcutaneous adipose	581	0	No qualifying IV	—	—
TSHR / Visceral adipose	469	1	0.50 (0.43–0.58)	4.09e-19	4.09e-18
IGF1R / Visceral adipose	469	0	No qualifying IV	—	—
TSHR / Whole blood	670	0	No qualifying IV	—	—
IGF1R / Whole blood	670	0	No qualifying IV	—	—

Table S3. Fixed-panel tissue extension with FinnGen GD as the primary disease dataset. IVs are retained instrumental variants. ORs are per unit of normalized, inverse-normal-transformed expression. Adjusted P = min(1,10P); the ten-comparison denominator includes unestimable cells. Not tested means the gene was absent from the tissue eGenes file; no qualifying IV means the minimum eQTL P did not meet 5×10^{-8}. All four estimable combinations used one SNP (F 32.1–61.1). The 40 secondary comparisons use a separate denominator of 40 and are retained, with all source statistics, in Tissue_extension. Cultured fibroblasts and adipose tissues are not orbital tissue. Original detection benchmarks remain in Detection_original.

Figure S1

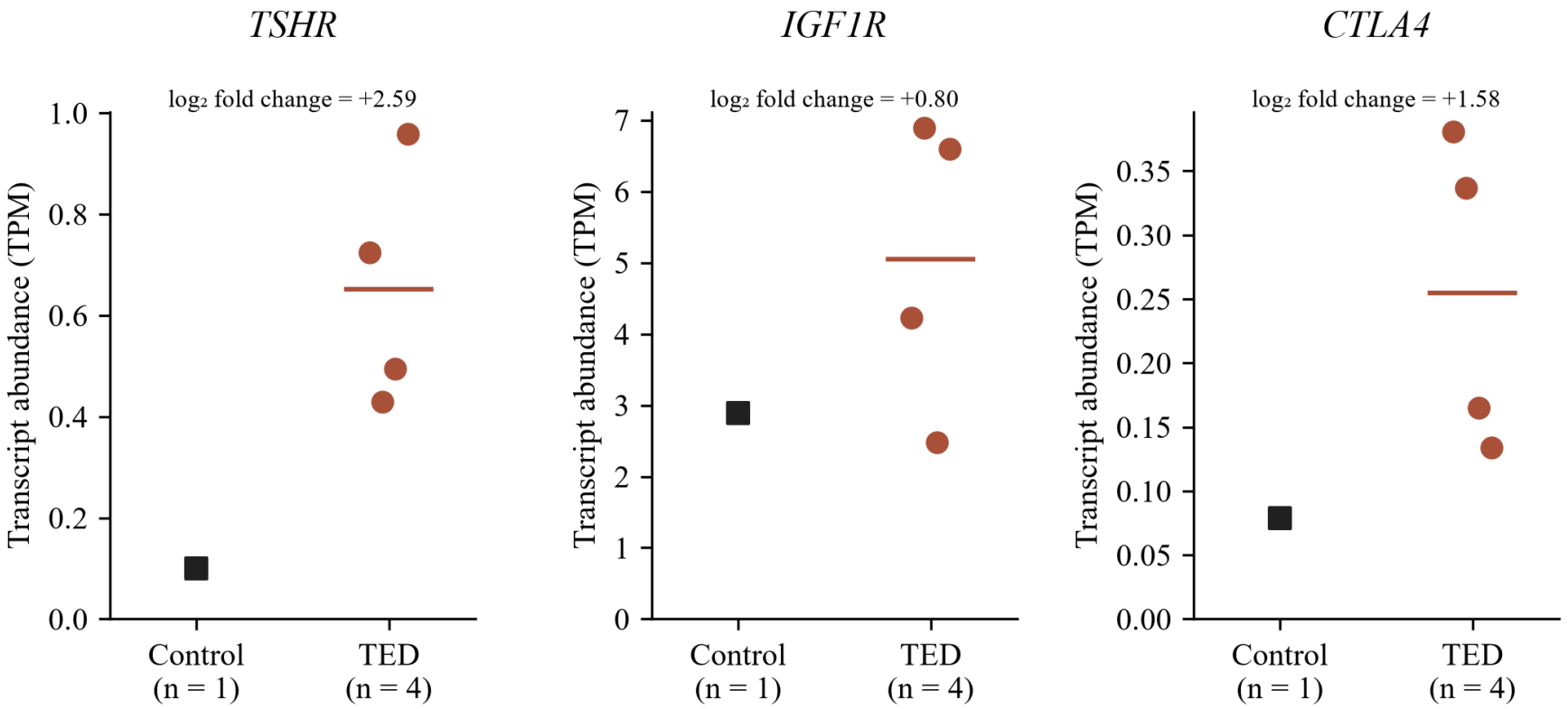


Figure S1. Descriptive orbital expression. Previously reviewed sample-level transcript abundance for TSHR, IGF1R and CTLA4 in four TED patients and one control. Points are biological samples; horizontal lines show TED means. Fold changes use log2[(mean TED TPM + 0.01)/(control TPM + 0.01)]. Axes differ by gene. No inferential expression test was performed; these observations do not validate the genetic findings. The figure is unchanged from the earlier review package.

Figure S2

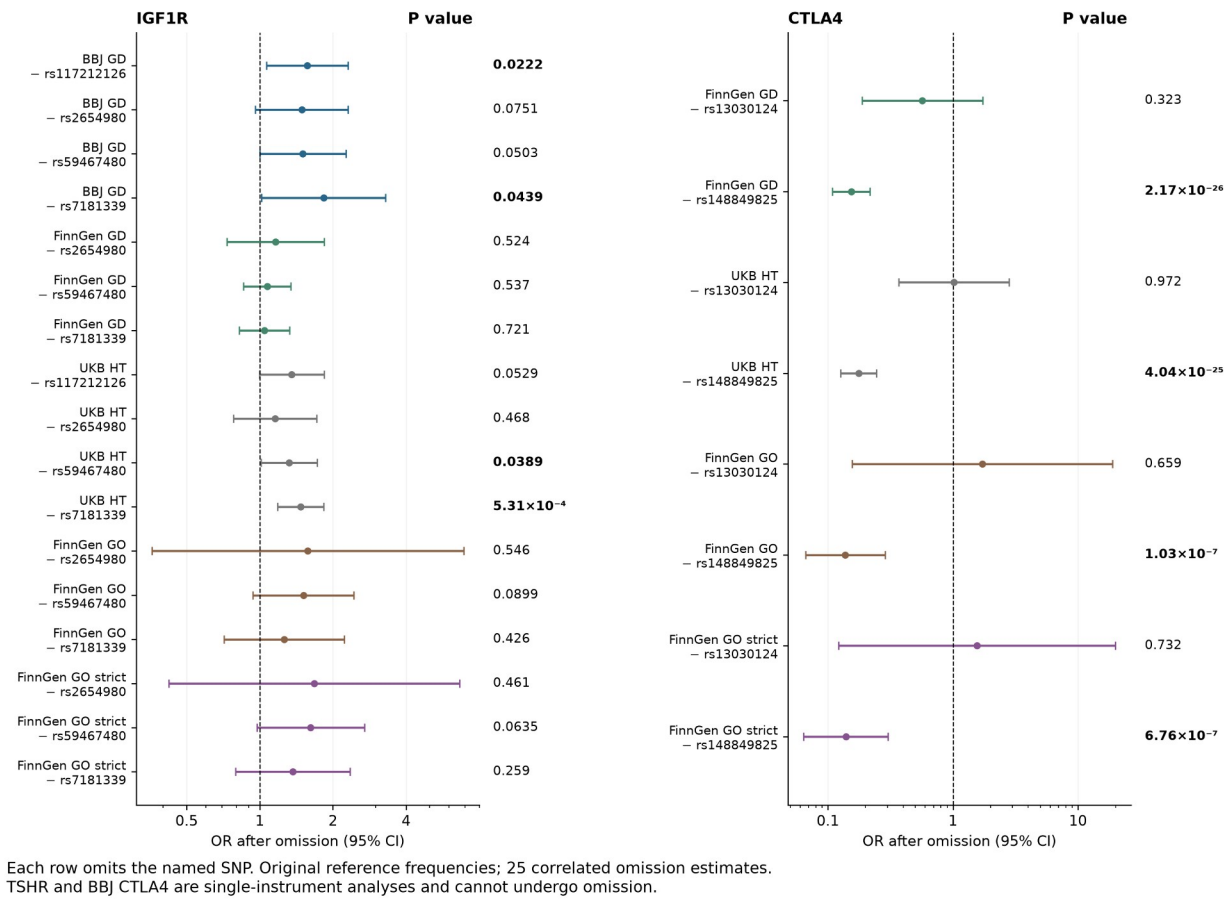


Figure S2. Instrument-omission sensitivity. Twenty-five reference-frequency SNP-omission estimates for IGF1R and CTLA4 across the five datasets. The named SNP is omitted; Wald ratios apply when one remains and IVW when at least two remain. Horizontal lines show 95% CIs; the vertical reference is OR = 1. Bold P values mark nominal $P < 0.05$ only. Original full-set estimates are in Table 2. TSHR and BBJ CTLA4 were ineligible because each has one instrument. These correlated post hoc diagnostics are not independent replication.